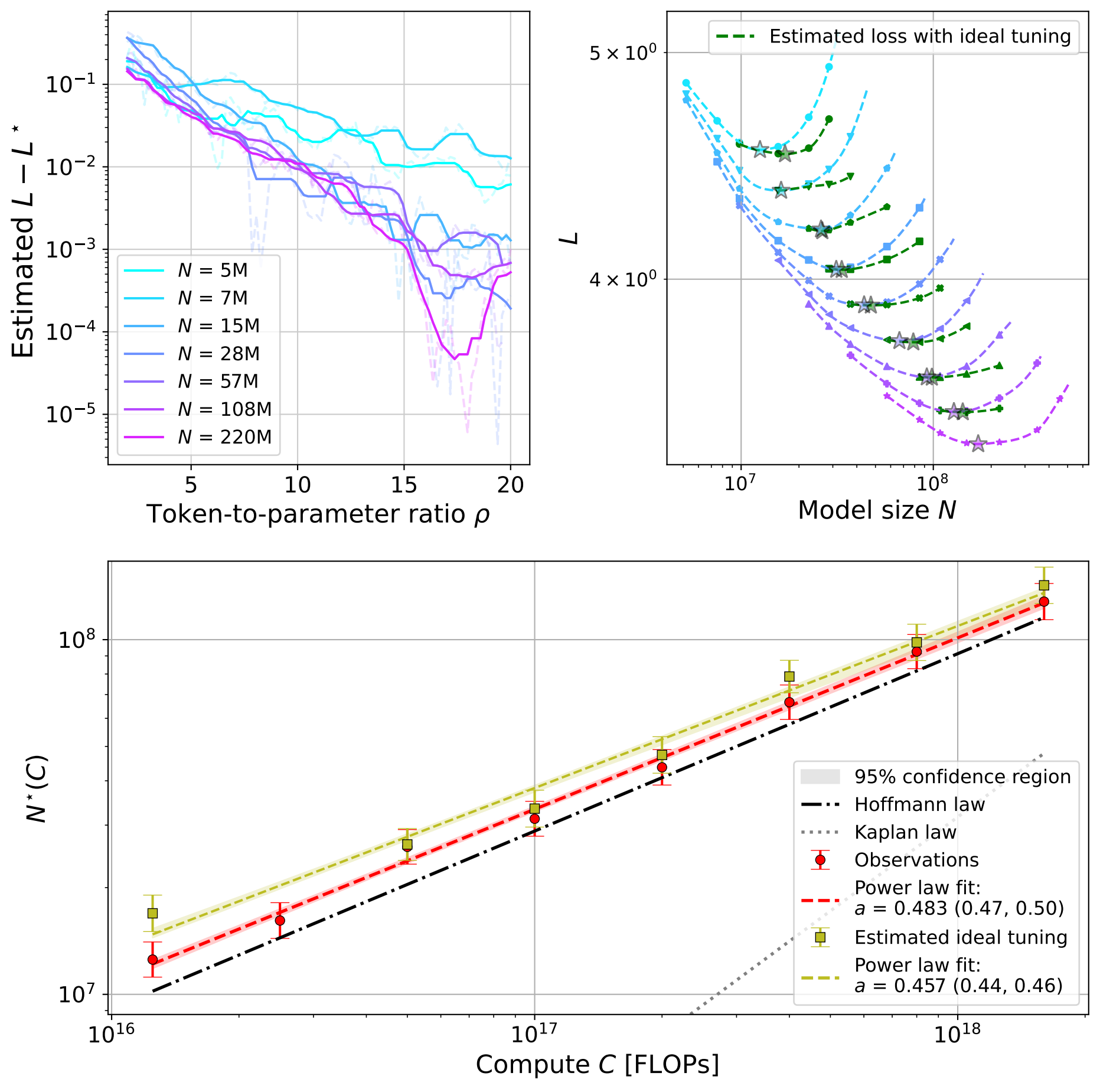

Estimated $L - L^\star$
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
N = 5M
N = 7M
N = 15M
N = 28M
N = 57M
N = 108M
N = 220M
5
10
15
20
Token-to-parameter ratio ρ
Estimated loss with ideal tuning
5×10^0
4×10^0
L
10^7
10^8
Model size N
$N^\star(C)$
10^8
10^7
95% confidence region
Hoffmann law
Kaplan law
Observations
Power law fit:
a = 0.483 (0.47, 0.50)
Estimated ideal tuning
Power law fit:
a = 0.457 (0.44, 0.46)
10^{16}
10^{17}
10^{18}
Compute C [FLOPs]